Reputational Shocks and Commitment Devices: Differential Effects on Foreign Direct Investment in Developing Economies

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Abstract

Why capital doesn't flow from rich to poor countries is a relevant but not at all new question. As such it has gotten many answers since it was first posted by Lucas in 1990. The literature has shown that institutional quality is a key factor in explaining this paradox. But how do investors find out about the quality of a country's institutions? Investor-State Dispute Settlement (ISDS), an institutional arrangement present in Bilateral Investment Treaties (BITs), should work as a mechanism where institutional quality gets revealed. Having a "case" would imply a negative reputational shock and the country's future FDI should diminish, even more, if the country loses it. Since most disputes occur in the context of a BIT, which functions as a commitment device, the overall result should also include this effect. I test this hypothesis and identify both effects by estimating a structural gravity equation using a Poisson pseudo maximum likelihood method. Results for my benchmark model show a reputational shock that reduces FDI inflows by USD 20 million one year after a dispute. This is more than compensated by the commitment device effect that increases these flows by USD 95 million. My analysis of heterogeneous effects shows that most negative results are driven by a subsample of countries with a large history of disputes. This shows that the reputational effects are small and that the interaction of the whole set of institutions promotes FDI in developing economies.

Keywords: Foreign Direct Investment, Investor-State Dispute Settlement, Reputation, Developing Economies.

JEL Codes: F21, F53, F55

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1 Introduction

Why capital doesn't flow from rich to poor countries is a very relevant but not at all new question. As such, it has gotten many answers since it was first posted thirty years ago when Lucas (1990) shows the data do not match the neoclassical prediction that countries with lower capital-to-labor ratio receive more capital inflows. Market imperfections, in particular the existence of a "political risk" due to the lack of mechanisms to enforce contracts, explain this paradox. Low institutional quality is the main reason for the lack of capital inflows in poorer economies, and those looking to attract capital inflows should reduce corruption, increase government stability, and strengthen the protection of property rights (Alfaro et al. (2008)).

Poorer countries try to overcome this lack of trust in their institutions through mechanisms that legally bond their behavior and give a positive signal to foreign investors. International investment agreements are developed for this purpose. These commitment devices increase the cost of not acting according to the rule of law. They aim to reduce the perceived risk of expropriation for multinational firms, and the cost of capital for countries hosting foreign investments (Sykes (2005)). They also give firms the right to pursue legal action for compensatory damages if a host country engages in unlawful acts.

How do investors find out about the quality of a country's institutions when deciding where to locate their investment? In this paper, I analyze the role of Investor-State Dispute Settlement (ISDS) an institutional arrangement that gives foreign direct investors the right to take disputes with host governments to international arbitration and is present in Bilateral Investment Treaties (BITs) and other international investment agreements. BITs are the most frequent of these instruments, defined by the United Nations Conference on Trade and Development as "an agreement between two countries to promote and protect investments made by investors from respective countries in each other's territory" (UNCTAD, 2019).

I argue that ISDS is a mechanism where institutional quality is revealed. The mere fact of having a "case" sends a bad signal to the investors and gives the country of destination a bad reputation. After having a case, a country's foreign direct investment is negatively impacted. However, most ISDS disputes are based on BITs, which are international commitment devices, hence the overall result includes both opposite effects.

I extend this initial proposition by considering two additional effects. One is losing the case, meaning the international arbitration panel rules that the country breached the firm's contract. This should have a stronger negative reputational effect on future and current investors. The second is the existence of a complementarity between domestic institutions and international arrangements. I analyze heterogeneous effects by identifying countries with a "low-frequency" of cases as those with a total number of cases since 1987 of less than the sample mean. My robustness check removes China from the sample as it is the developing economy with the largest FDI amounts in the analysis period.

I use a gravity equation to investigate the reputational effect of ISDS and the commitment device effect of BITs on FDI flows and positions in developing economies. I estimate a multiplicative model of bilateral FDI using a Poisson pseudo maximum likelihood method (PPML) (Santos Silva and Tanreyro (2006)). I include fixed effects for the country of origin-year and country of destination-year to account for time-varying GDPs and multilateral price terms, and country pair fixed effects to control for endogenous characteristics of both countries that could impact the outcome of interest (Baier and Bergstrand (2007)).

I use several publicly available sources to construct my dataset which I arrange in a dyad form. I construct a bilateral dataset of inward direct investment positions with information from the International Monetary Fund's Coordinated Direct Investment Survey (CDIS). This source covers the years from 2009 to 2015 determining my analysis period. I obtain information about all the BITs available from UNCTAD's International Investment Agreements Navigator. Finally, the data on ISDS cases comes Wellhausen (2015) and provides information on international investment arbitrations filed by foreign investors against states from 1987 to 2015.

Findings in my benchmark model suggest the overall effects of these international arrangements are positive and increase FDI inflows and positions in developing economies. There is variation in magnitudes, which are larger two years after a case, particularly for positions. When considering the composition of these overall positive effects, I find a negative reputational shock of 21 and 27 million dollars in bilateral FDI inflows one and two years after the case respectively. This is more than compensated by the positive effect of the interaction between a BIT and a case of 95 and 192 million dollars for each specification. Similarly, the negative effect of a case in FDI positions is about 65 million dollars while the interaction effect is 117 million dollars with a one-year lag and of 185 with a two-year lag.

Both extensions to the benchmark model yield similar results. The first one includes the effects of losing the case. These results show a puzzling positive effect the first year after a case which is the largest of all the estimations. Nevertheless, in line with previous findings in the literature, I find a negative economic impact when considering FDI flows two years after the case, and FDI positions one year later. The second extension, the interaction of international and domestic institutions, shows similar results though different in magnitude. I find that the complementarity effect is the largest for FDI positions two years after a case, which is the most stable specification throughout models in terms of sign and magnitude.

The analysis of heterogeneous effects shows positive economic impacts in FDI flows and positions larger for the subsample of countries without a history of international disputes than for the whole sample. This suggests ISDS history matters for investors and there is no reputational shock for countries that do not have a large track of international arbitration with firms. A robustness check estimating the benchmark model removing China as an economy of destination finds similar positive effects in most specifications. However, for FDI positions one year after the case results are negative, as in the model extensions. This suggests the reason the estimations for the benchmark model are not negative is a large positive China effect.

In sum, the negative reputational effect associated with an ISDS case is larger than the commitment device effect of a BIT for FDI inflows two years after the developing economy of the destination loses the case, and when it interacts with the quality of the domestic institutions. For FDI positions this negative shock occurs one year after the case in all but the benchmark model. These effects are mostly driven by countries with a large history of international disputes showing that the mere presence of a case does not provide a bad reputation but rather a large track of them. In all models, negative results disappear two years after the dispute when considering the more stable measure of FDI bilateral positions.

This paper contributes to the multidisciplinary studies that analyze the effects of investor-state dispute settlement on developing economies' reputations through its impact on foreign direct investment. I extend this literature by developing a gravity equation to identify the effects of BITs, cases, and their interaction. This is relevant since both legal arrangements usually go together. In addition, using a PPML estimator overcomes many of the empirical challenges from the characteristics of FDI data such as heteroskedasticity, skewness, and the large presence of zeros.

Directly related to my paper, Allee and Peinhardt (2011) claims that the mere appearance of a government in international arbitration sets a negative shock to the country's reputation and discourages future FDI flow. Thus, BITs can increase foreign direct investment only if countries are not taken before an arbitration panel. If this happens, future flows of FDI diminish, even more, if the country loses the case. I extend this analysis by accounting for country pair flows and positions and developing a different estimation framework.

Similarly, Aisbett et al. (2018), using OLS to estimate a gravity equation for FDI flows in developing economies from 1980 to 2010, find that BITs signed before the host country faces a dispute have a positive impact on bilateral FDI flows. After a dispute, FDI flows from BIT partners decrease more strongly than those from countries without this agreement. Furthermore, after a host country faces a dispute, the net impact on bilateral FDI flows from signing a new BIT is nearly zero. Their treatment of the "claim" variable is different from the one I use in this paper since it does not take into account which is the home country of the investor placing the dispute.

Host governments develop innovative institutions such as BITs to overcome political risks that discourage foreign investors. Tobin and Rose-Ackerman (2011) find these mechanisms do not substitute for weak local institutions, rather they complement each other. FDI reflects the investors' confidence in a country and BITs create commitments from host countries to an appropriate environment for foreign investment. They identify two channels for this: one is the financial costs that a country faces for violating the agreements. The other, which directly relates to the main topic of this paper, is that countries that violate a BIT face reputational costs that discourage future investors.

My use of a gravity equation to estimate the effect of BITs on bilateral FDI flows relates to Falvey and Foster-McGregor (2017) and their finding of non-linear effects. They show that the effectiveness of BITs on FDI is increasing in the difference in GDP (size) and GDP per capita (development) between the source and host countries. They argue this finding is consistent with the idea that BITs are more effective in promoting foreign direct investment when source and host countries have large institutional differences. The authors also find evidence supporting a complementary relationship between BITs and domestic institutions.

However, Berger et al. (2013) argue that the literature is not conclusive about the role of BITs in promoting FDI because it treats them as "black boxes", considering them as homogeneous while they are diverse. Part of this diversity reflects the modalities of the admission phase of FDI and dispute settlement mechanisms. This second factor refers to

ISDS, a credible commitment from the home country against discriminatory measures to foreign investors after they are established in the country. The authors find this plays a minor role and that investors react more to guarantees in the admission phase before they are established in the country. Hence to promote FDI, governments in developing economies should provide legal security and predictability in access conditions rather than ex-post mechanisms.

The results from this paper could inform the current efforts to reform ISDS that are taking place at the United Nations Commission on International Trade Law (UNCITRAL) and help develop a mechanism that is useful for firms and countries. Further research should incorporate another set of costs that ISDS has for countries. In particular, I do not consider the cost of awards, which have been increasing later making a case by legal authors against "crippling compensations". Similarly, I do not include the "regulatory chill" effects by which some governments stop implementing domestic regulations to prevent disputes.

The rest of this paper is organized as follows: in section 2 I present background information on ISDS cases and venues, and in section 3 the economic framework I use for my analysis. In section 4 I explain the construction of the dataset and main variables of interest, and in section 5 my empirical framework and estimation method. In section 6 I present and discuss my results, and in section 7 the analysis of heterogeneous effects and robustness checks. I conclude in section 8.

2 Background on Investor-state Dispute Settlement

Investor-state dispute settlement procedures present in international investment agreements allow multinational firms to access arbitration panels that decide if the host state incurred a breach of contract or any other aspect of a treaty. ISDS has specific characteristics that differentiate it from trade dispute settlement procedures, as they are designed to solve different types of problems (Ossa et al. (2020)).

The economic motivation for ISDS is the holdup problem present in international investments (Stahler (2021)). If the investment is country-specific and there is a lack of complete contracts, investors foresee this issue and the host country may face underinvestment. While models for the holdup problem usually refer to a firm-to-firm activity, the fact that in this setting one of the actors is the government makes this a particular mechanism.

ISDS is a controversial tool. It can be seen as a good mechanism to promote FDI in poorer countries. Since its role is to guarantee the rule of law and give investors confidence against expropriations, it might attract investment in developing economies whose institutions may not be as reliable to investors from richer countries.

In a 2020 survey about investors' perceptions of ISDS, investors answered that they are mostly satisfied with the current system (Queen Mary University of London (2020). Furthermore, they prefer it to other mechanisms such as government negotiation or litigation in the host state's courts. Investors also claim that their investment decisions are positively influenced by the stability and predictability of the host state's legal framework. At the same time, the availability of ISDS and the host country's history in investor-state disputes also influence their investment decisions.

However, other actors see ISDS as a mechanism undermining a country's sovereignty. This was clearly expressed in October 2017 when 230 Law and Economics professors sent a public letter to former president Trump urging him to remove ISDS from NAFTA (now the United States–Mexico–Canada Agreement) and to avoid including such mechanisms in any future trade or investment treaty. They claimed that "The inclusion of ISDS in U.S. trade and investment deals (...) threatens to dilute constitutional protections, weaken the judicial branch, and outsource our domestic legal system to a system of private arbitration that is isolated from essential checks and balances."[1] This characteristic of ISDS is particularly relevant in areas such as public health or the environment. By taking a lax definition of what constitutes an expropriation, many of these policies can be challenged by foreign firms.

2.1 Venues

There are several venues for international arbitration between states and firms. Some of them are the United Nations Commission on International Trade Law (UNCITRAL); the Permanent Court of Arbitration (PCA) in The Hague, Netherlands; and the Arbitration Institute of the Stockholm Chamber of Commerce (SCC). However, the most widely used venue is the International Centre for Settlement of Investment Disputes (ICSID) from the

[1] Available at here

World Bank.[2]

Its goal is to provide an impartial international venue for resolving disputes between states and foreign investors. It was established in 1966 by the ICSID Convention, and the first case was registered in 1972. It has more than 160 signatory and contracting states. Each member state participates in the administrative council, which elects the secretary-general in charge of the daily operations. Being a member of this institution can be one of the most effective bonding mechanisms for a developing economy since not respecting an ICSID decision can have reputational consequences for countries and possibly compromise future support from the World Bank group.

The geographical distribution of these cases shows that less developed economic regions like East Europe and South America have more cases than more developed ones like North America. This fact goes in line with the argument that foreign investors in economies with weaker institutions, like the transition economies in East Europe, are those who benefit the most from the existence of ISDS. ICSID also reports that the tribunal decided 66% of the disputes in 2019, while 34% were settled or discontinued.

2.2 An Illustrative Case

An interesting is the case "Phillip Morris v. Uruguay".[3] In 2010 the multinational tobacco company Phillip Morris failed a dispute against Uruguay based on the Switzerland-Uruguay BIT. The firm alleged that Uruguay violated the bilateral investment treaty through regulatory measures in the tobacco industry and demanded a compensation of 25 million dollars. These measures were part of an important series of anti-tobacco regulations that the Uruguayan government had promoted since 2005 and made Uruguay the first Latin American country to be smoke-free in public spaces (Green (2021)).

Phillip Morris challenged the adoption of a single presentation requirement, obliging tobacco manufacturers to sell only one variant of cigarette per brand family, and an 80% increase in the size of graphic health warnings on cigarette packages. The allegation was that, by banning many of the cigarette variants and substantially diminishing the value of the remaining ones, Uruguay was expropriating the firm's brand assets, including intellectual property, without compensation.

[2] See ICSID

[3] More information here

The dispute won international recognition for being the first case of its kind and having the potential to set jurisprudence. The World Health Organization, the Pan American Health Organization, MERCOSUR, and many anti-tobacco organizations supported Uruguay's position.

After six years, ICSID dismissed Phillip Morris' claims in a widely celebrated resolution by the international public health community. This case is relevant for this paper because it is hard to guess if this dispute would generate a negative reputational effect. Is the presence of an ISDS case per se a bad reputational shock?

3 Economic Framework

3.1 Structural Gravity Equation and FDI

With the development of the theoretical foundations of the gravity equation, the structural gravity model was developed and its use became suitable for other types of flows, such as FDI. Head and Ries (2008) develop a gravity theoretical equation for FDI as acquisitions motivated by international corporate control. Head and Mayer (2021) argue that modern gravity models allow estimating the underlying costs of cross-border movement for different types of flows where each type is the outcome of a discrete choice problem. For capital, they argue is an asset owner looking to sell to the highest bidder. In this paper, I follow an approach similar to the one used to evaluate the economic impact of Economic Integration Agreements (EIA), but for the case of BITs.

3.2 Conceptual Approach

Developing countries that want to promote inward FDI sign BITs as a binding mechanism. In this way, they become "institutionally closer" to the richer economies from where they want to attract investment. Later, the presence of a dispute makes those two countries become "institutionally more distant". This can be seen in stages.

In the first stage, a developing economy (A) wanting to increase its inward FDI signs a BIT with a richer country (C) to signal a commitment to the rule of law. A profit-maximizing firm from country C deciding to invest in a developing economy analyzes the costs of investing in A and another similar country, B. Country B does not have a BIT with country C. In this framework the main cost is the risk of expropriation, which is lower in country A due to the BIT, so the firm decides to invest there. Holding everything else equal, the hypothesis suggests that investment from country C should go more to country A than to B.

In the second stage, a multinational firm from country C files an international dispute against country A through the ISDS instrument in the BIT. This is a revealing mechanism of the true type of institutions present in country A. It signals to the rest of the investors that the country in question might have committed an unlawful act against the foreign firm. This increases the cost of investment through the risk of expropriation. If the dispute is resolved in favor of the firm, it explicitly reveals the "bad type" of the host country's institutions. Investors identify such cases as a negative signal regarding property rights and stability in that country and should be less willing to invest or might even leave the country.

Hence, whenever two countries sign a BIT, most of which include ISDS, they are signaling their willingness to abide by its regulations. However, the real behavior of the country and whether it is the type that respects contracts will not be revealed until an investment dispute arises against the sovereign.

ISDS and BIT operate in two ways for developing economies. First, it reduces expropriation risk when the treaty is in force making the two economies closer through a commitment device effect. When there is a dispute, the perception of expropriation risk increases for other firms from the investor's home country. This is a negative reputational shock that makes the two economies more distant. Therefore we should expect fewer firms to be willing to invest there, even some firms disinvesting, and expected FDI inflows and positions to diminish. This can be expressed in a gravity equation form:

$$E[F_{do}] = exp[C_{do} * R_d - D_{do}]M_d S_o \tag{1}$$

Where $E[F_{do}]$ is expected foreign direct investment; C_{do} is the commitment device between the country of destination d and country of origin o; R_d is the reputational shock on the country of destination. These are the main parameters to estimate. As is usual in gravity models, this expression means that expected FDI in the country of destination from the country of origin depends negatively on the distance between the two countries $-D_{do}$, and on the sizes of economies of destination and of origin M_d, S_o.

4 Data

In this section, I describe the different publicly available sources I use to build my dataset. I arrange all the data in a dyad form to include the relevant variables for each country pair in the sample.

4.1 Bilateral FDI

Using the International Monetary Fund's Coordinated Direct Investment Survey (CDIS), I construct a bilateral dataset of inward direct investment positions. I take the difference in the yearly positions within each country pair for the FDI flow variable.

The survey compiles positions by the immediate direct investor, by counterpart economy, for net equity and net debt instruments, by the end of December of each year (IMF (2015)). It constitutes direct investment when an investor residing in one economy has 10% or more of the voting power in a company resident in another economy. This operationalizes the concept of having "control or a significant degree of influence" on the firm management.

The dataset contains levels of FDI equal to zero when this is the reported value, negative values, and missing information (including data reported as confidential). The period covers the years from 2009 to 2015.

I define developing economies as those not classified as high-income in 2009 by the World Bank Analytical Classifications presented in WorldBank (2021). Operationally, this means leaving out of the IMF sample destination countries that had a GNI per capita greater than USD 12,195 in 2009.[4]

Hence, this leaves 165 economies of destination and 126 economies of origin (full list in Table A6 in the Appendix).

Figure 1 reflects that, during the period covered by the sample, FDI inward positions in developing economies fluctuate. In the beginning, there is a sharp increase following the 2008 financial crisis. This however diminishes from 2010 to 2012, after which it starts growing again. The final years show a relative stagnation in our main variable of interest.

[4]A few countries that changed category during the period and became classified as high-income are part of the sample

4.2 Bilateral Investment Treaties

I obtain information about all the BITs available, which are more than 3000, from *International Investment Agreements Navigator | UNCTAD Investment Policy Hub* (2021). Since the data need to be gravity-type, each of the two countries signing the BIT is included both as country of destination and country of origin. The source provides two dates, one when the treaty was signed and the other when it was enforced. I use this second date for the estimation.

Table 2 shows the number of BITs in force for the 10 developing economies in the sample with more BITs in 2015. It reflects that these economies are recurring to bilateral investment treaties as an active form to foster foreign investment in their economy.

4.3 Cases

I use publicly available data on Investor-State Dispute Settlement (ISDS) from Wellhausen (2015). The dataset provides information on international investment arbitrations filed by foreign investors against states from 1987 to 2015.

I merge it with my previous data and create the variable "case" which equals 1 for each year a country pair has a case, and zero otherwise. I obtain a total of 90 cases.[5] Of those cases, 81 were between a country pair that also had a BIT in force that year, the rest of the cases were based on other types of international investment agreements.

Some features of the data are relevant for the analysis in this paper. The same number of cases in the sample were won by investors than by states (44), with 2 cases being settled. The most represented economic sectors in the disputes are Mining (20%), Manufacturing (18%), Utilities (14%), and Oil (10%). I make a sub-classification of countries defining as "high offenders" those that had more than the average number of cases considering the information from 1987 to 2015. Countries that had 9 or more ISDS cases are included in this category. Finally, the most used venue for the cases in the sample is ICSID (66 cases), followed by PCA (11) and UNCITRAL (6).

Considering the income level of the country pairs involved in terms of the World Bank classification, in 90% the country of origin of the investment is a rich economy. The great

[5]This is one case per year and country pair, except for El Salvador and the US which have two cases in 2009, Turkmenistan and Turkey with two cases in 2010, and Venezuela and The Netherlands three cases in 2011.

majority (92%) of the destination economies are middle-income (either Upper or Lower). The full information is in Table A5 in the Appendix.

4.4 Domestic Institutions

To test Tobin and Rose-Ackerman (2011) proposition that BITs do not substitute for weak local regulation but rather complement them, I include a variable to capture this local institutional effect.

From the Doing Business project (WorldBank (2020)), I get the variable *Score-enforcing contracts* (DB04-15 methodology) that captures the procedures, time, and cost for resolving a commercial dispute through a local first-instance court. It measures the gap between an economy's performance and the regulatory best practice on the Enforcing Contracts indicator components. It is calculated as the simple average of the scores for Time (days), Cost (% of claim value), and Procedures (number). The score ranges from 0 to 100, where 0 represents the worst regulatory performance and 100 the best regulatory performance.

I construct a dummy variable *Domestic* that equals one if the country of destination has a score equal to or larger than the sample mean plus one standard deviation of *Score-enforcing contracts*, and zero otherwise. Thus, countries having an indicator of one in this variable rank high for enforcing contracts, reflecting a good quality of domestic institutions.

4.5 Summary Statistics

The summary statistics for the dataset I constructed as described above are displayed in Table 3. This represents all FDI inward positions from 2009 to 2015 with "non-rich" economies as destination countries. The 139,216 observations represent 19,888 undirected country pairs through the seven years in the sample.

5 Empirical Framework and Estimation Method

I use a gravity equation to investigate the reputational shock of ISDS and the commitment device effect of BITs, on FDI positions and flows in developing economies. As proposed Baier and Bergstrand (2007), I include fixed effects for country of origin-year, country of destination-year, and country pair. The first two types of fixed effects account for time-varying GDPs and multilateral price terms. The third type controls for unobservable time-invariant characteristics between the country pair.

The authors use this approach to estimate the effects of Economic Integration Agreements and prove that, given that its determinants are slow-moving, estimations through gravity equations could use panel techniques and data to avoid endogeneity bias and find the correct Average Treatment Effects (ATE). I argue that the same is true for BITs and this estimation strategy is also useful in my specification.

I use a double interaction between the country pair having a case and BIT to capture my main hypothesis that both mechanisms have simultaneous effects. I obtain the effect of cases (treatment) on country pairs with BITs (treated), in a similar way as Baier et al. (2018) introduce interaction terms to estimate Average Treatment Effects on the Treated (ATT) differentiating treatments and types of treated countries.

I follow the Santos Silva and Tanreyro (2006) approach to estimate gravity equations and other multiplicative models. The authors claim that when these models are log-linearized and estimated by OLS, the parameters are biased estimations of elasticities in the presence of heteroskedasticity. This is an implication of Jensen's inequality that states $E[ln(y)] \neq ln[E(y)]$. To overcome this, they suggest estimating the models in their multiplicative form using a Poisson pseudo-maximum-likelihood (PPML) estimation method.

PPML addresses the issue of heteroskedasticity and the presence of 0 values in the dependent variable, which is very common in FDI data. With this specification, the data does not need to be Poisson. The only requirement for the consistency of the estimator is that the mean is correctly specified: $E[y_i/x] = exp(x_i\beta)$. Still, inference must be performed based on robust standard errors since the estimator does not account for all heteroskedasticity.

In addition, Fally (2015) finds that estimating gravity equations using the PPML estimator with fixed effects is consistent with the introduction of the "multilateral resistance" terms proposed by Anderson and Wincoop (2003).

5.1 Benchmark model: Reputation and Commitment

I use a PPML estimation method with high dimensional fixed effects (Correia et al. (2019)). This specification only allows for non-negative numbers on the left-hand side, therefore negative inward positions (liabilities) are not included. The gravity equation to be estimated corresponds to the following econometric specification:

$$FDI_{do,t} = exp[\beta_1 BIT_{do,t} CASE_{do,t-n} + \beta_2 BIT_{do,t} + \beta_3 CASE_{do,t-n} + \gamma_{dt} + \delta_{ot} + \alpha_{do}]\varepsilon_{od,t}. \tag{2}$$

Where $FDI_{do,t}$ represents the bilateral FDI inward position or flow (in current million dollars) into destination country d from origin country o in year t; $BIT_{do,t}$ equals to 1 all the years there is a BIT in force between the country-pair and zero otherwise; $CASE_{do,t-n}$ equals to 1 if there is a case between the country pair in the previous n years (n=1, 2) and zero otherwise; γ_{dt} is country of destination-year fixed effect; δ_{ot} is country of origin-year fixed effect; α_{do} is the country pair fixed effect; and $\varepsilon_{od,t}$ is the error term. The economic motivation for including one and two-year lags is that the potential effects of a reputational shock are less likely to materialize immediately.

5.2 Losing a Case

The assumption of an ISDS case revealing the destination country's institutional characteristics relies on the fact that cases are resolved by an independent panel that analyzes the dispute impartially. If this panel finds that the investor was correct in its claim, meaning that the country loses the dispute, it should have a stronger negative shock to the country's reputation, as found by Allee and Peinhardt (2011). Thus, negative results from cases have information that, when revealed to the market, may deteriorate a country's reputation beyond just having a case.

I test for this by including the indicator *Lost* as a third interaction into the benchmark model of equation 2. This variable equals 1 if the investor wins the dispute and zero otherwise.

I estimate the following equation.

$$\begin{aligned} FDI_{do,t} = & exp[\beta_1 BIT_{do,t} CASE_{do,t-1} Lost_{d,t-1} + \beta_2 BIT_{do,t} + \beta_3 CASE_{do,t-1} + \\ & \beta_4 Lost_{d,t-1} + \gamma_{dt} + \delta_{ot} + \alpha_{do}]\varepsilon_{od,t}. \end{aligned} \tag{3}$$

5.3 Domestic Institutions Complementarity

A complementarity with domestic institutions implies that BITs successfully attract foreign investors to poorer countries when interacting with good-quality domestic regulations. Both types of institutions interact, and the international commitment device does not substitute poor quality in local laws (Tobin and Rose-Ackerman (2011)).

I extend my benchmark model to test this hypothesis. I modify equation 2 to introduce a triple interaction with a variable that reflects the quality of the domestic regulations for investors. The indicator *Domestic* equals 1 if the country of destination ranks above the sample mean plus one standard deviation in its score for enforcing contracts as measured by the indicators collected by the World Bank, and zero otherwise.

$$\begin{aligned} FDI_{do,t} =& exp[\beta_1 BIT_{do,t} CASE_{do,t-1} Domestic_{d,t} + \beta_2 BIT_{do,t} + \\ & \beta_3 CASE_{do,t-1} + \beta_4 Domestic_{d,t} + \gamma_{dt} + \delta_{ot} + \alpha_{do}]\varepsilon_{od,t}. \end{aligned} \tag{4}$$

6 Results and Discussion

In this section, I present and discuss my results, estimated as described in Section 5. First for my benchmark model, then for losing a case, and finally for the complementarity of international and domestic institutions. The estimated coefficients can be converted into semi-elasticities by computing $(e^{\hat{\beta}_i} - 1)$*100. I apply this conversion to the coefficients that are statistically different from zero and then evaluate them at the mean FDI of the sample to obtain the economic impact.

6.1 Commitment Devices Compensate Reputational Shocks

I test my hypothesis of the presence of two opposed effects in ISDS and BITs: a negative reputational shock after a dispute, and a positive effect given by the commitment device that is the international agreement. Table 4 shows the estimated coefficients for my benchmark model in Panel A, and the overall effects and economic interpretation in Panel B.

The reputational shock is captured by *Country pair has a case*, a dummy variable that equals to one all the years after there is an ISDS case between a country pair. The variable is lagged by one and two years after the dispute. The variable *BIT* captures the commitment device. It equals one if a bilateral investment treaty is in force between the country pair and zero otherwise. The overall effect is captured by the interaction between the two variables plus the individual effects.

Columns 1 and 2 display the estimation results for FDI flows. The first one includes the case variable lagged one period, and in the second it is lagged two years. Similarly, columns 3 and 4 show results for FDI positions between the country pair, after one and two years of having a case respectively.

In column 1 of Panel A, the coefficient on the BIT indicator is not statistically significant. The coefficient of the existence of a case between the country pair in the previous year is negative and statistically significant at 10% level, with a point estimate of -1.180. The semi-elasticity for this reputational effect is -69%, which evaluated at the sample mean represents a 21 million dollar negative impact in FDI inflows. The estimated coefficient for the interaction of the two variables is 1.435 and statistically significant at the 5% level. The semi-elasticity is 320% meaning a positive economic effect of USD 95 million.

In Panel B, I present the overall effects by adding the three coefficients. The total coefficient is 0.255, which converts into a semi-elasticity of 29%. This means that the negative effect of the reputational shock gets more than compensated by the commitment device effect of the bilateral investment treaty. Evaluated at the sample mean of USD 30 million, it represents an increase of 9 million dollars in FDI flows one year after having an ISDS case.

It is important to notice that because this is an exponential model, the sum of the effects of each coefficient considered separately does not equal the effect of the coefficients considered together. In my framework, the overall effect is the most important since it captures the interaction between the commitment device and the reputational shock.

In column 2 of Panel A, the coefficient for BIT is positive and statistically significant at the 10% level, with a point estimate of 1.141. This is a semi-elasticity of 213%, with an economic impact of 63 million dollars in FDI inflows. The coefficient for the country-pair having an ISDS case two years prior is -2.320, significant at the 5% level. This semi-elasticity of -90% is equivalent to a negative economic shock of 27 million dollars, slightly larger than the one-year lag. The interaction of these two variables estimates a coefficient of 2.010 with a significance level of 10%. As in the previous column, the interaction is also positive showing that the existence of a BIT compensates for the negative effect of the case.

The average treatment effects on the treated reflected in Panel B show an overall coefficient of 0.831. This equals a semi-elasticity of 130% and a positive economic impact of 39 million dollars on FDI flows two years after the existence of a case between the country of destination and a firm from the country of origin.

In column 3 of Panel A, we see no statistically significant effects for the BIT variable. The case variable lagged one year yields a point estimate of -0.498 with a 10% statistical significance, this represents a -39% effect on FDI positions, equivalent to a 65 million dollars loss one year after the case. Meanwhile, the interaction coefficient is 0.536, also statistically significant at the 10% level. This semi-elasticity of 71% represents a positive effect of 117 million dollars. These estimations show less statistical power than flows since they are significant at the 10% level.

The overall effects in Panel B show the sum of coefficients is 0.038, which represents a semi-elasticity of 4%, and an economic impact of 6 million dollars, evaluated at the sample mean for FDI positions of 165.65 million dollars.

Finally, column 4 in Panel A shows the coefficient for the BIT variable is not statistically different from zero. For the case variable lagged two years, the point estimate is -0.490, with a statistical significance of 10%. The semi-elasticity of -39% represents an economic loss of 64 million dollars in bilateral FDI positions two years after the country has a case. The coefficient for the interaction of the two variables is 0.749, which is a 111% semi-elasticity and 185 million dollar impact.

In column 4 of Panel B, the overall effects for positions two years after a case show a coefficient of 0.259, a 30% semi-elasticity, and an economic impact of USD 49 million.

In sum, the sole existence of a BIT might have a positive impact on FDI flows but not on positions in destination developing economies. Having a case between the country pair shows a negative effect on flows and positions, which is larger for flows as they are more vulnerable to changes in the economic environment. The largest effects occur on FDI inflows two years after having a case. This suggests a reputational shock on developing economies that face an international dispute with a foreign firm that takes time to materialize. However, the interaction term and the sum of all the effects are positive in all specifications, reflecting that an international commitment device such as a bilateral investment treaty compensates for the negative reputational effect of an international dispute.

These results might also reflect that investors trust the system and that ISDS is positive for them, as reflected in the survey by Queen Mary University of London (2020). Even though a case might signal a bad reputation for the country of destination (as the estimated coefficients show), the overall protection given by the international regulation system seems to be satisfactory for investors' decisions.

6.2 Negative Effects of Losing the Case

I estimate equation 3 to study if the negative reputational effect of an ISDS dispute does not come from the mere fact of having a case but rather from losing it. To facilitate the economic interpretation of the estimations, in Panel A of Table 5, I present the average treatment effects on the treated with the overall impact and leave the full set of estimations in Table A1 in the Appendix.

In Panel A, column 1 of Table 5, we see the effects for FDI flows one year after the country pair has a case. The sum of all the statistically significant coefficients is 2.055, which translates into a semi-elasticity of 681%. This represents an economic impact of 203 million dollars. So in this specification, we see an economic effect that is even larger than the USD 9 million from the benchmark model.

In column 2, the overall effects are negative, The sum of coefficients is -0.423, representing a semi-elasticity of -34%, and an economic loss of 10 million dollars. We see that reputational effects negatively affect FDI inflows in a developing economy one year after losing an international investment dispute with a foreign investor.

Column 3 shows the results for FDi positions one year after the country-pair has a case. Here we also see negative effects with the sum of the coefficients being -0.503, a semi-elasticity of -40%, and an economic loss of 65 million dollars. This result could represent disinvesting from the company's country of origin into the destination-developing economy as a result of the investor winning the international dispute.

Finally, column 4 reflects the overall results on FDI positions two years after the country-pair faces an international dispute. The coefficients sum to 0.250, which represents a semi-elasticity of 28%, and an economic impact of 47 million dollars. These results are similar to the same specification in the benchmark model, reflecting that after two years the reputational effect of having a case is the same as losing it.

In sum, I find evidence of a negative reputational effect on FDI inflows in a developing economy two years after losing a case, and on FDI positions after one year. When considering the estimations for each variable in Table A1, the sole fact of an investor winning a case does not provide enough information for a reputational effect. The interaction of the whole set of variables, that represent different institutional devices, provides a clearer picture of the developing economy of a destination's reputation for investors.

6.3 Domestic Institutions Complement International Devices

I estimate equation 4 to study if there is a complementarity effect between bilateral investment treaties and the quality of domestic institutions. In Panel B of Table 5, I present the average treatment effects on the treated with the overall impact and the full set of estimations in Table A2 in the Appendix.

In Panel B, column 1 of Table 5, we see the effects for FDI flows one year after the country pair has a case. The sum of all the statistically significant coefficients is 0.234, which converts into a semi-elasticity of 26%. This represents an economic impact of 8 million dollars, similar to the USD 9 million from the benchmark model.

In column 2, I show the results for FDI flows two years after the country pair has a case. The overall effects are negative, with the sum of coefficients being -2.338, representing a semi-elasticity of -90%, and an economic loss of 27 million dollars. Two years after having a case, its interaction with the quality of domestic institutions reflects a negative reputational effect, similar to what I find when the destination country loses the case but in a larger magnitude.

Moving into the impact on bilateral FDI positions, column 3 shows negative effects one year after having a case. The sum of all non-statistically zero coefficients is -0.729, with a semi-elasticity of -52%, and an economic loss of USD 86 million. These results are also the same in sign but larger in magnitude than those I find for losing a case.

Finally, column 4 reflects the results on FDI positions two years after having a case. In this specification, the results are positive, as in the analysis for losing a case. The total coefficient is 0.453, with a semi-elasticity of 57%, and an economic impact of 95 million dollars.

Overall, the interaction of international and domestic institutions provides a similar negative result to the investor winning the dispute. Except for the impact on flows one year after the case, domestic institutions' quality parallels the reputational shock of losing a case providing an effect of the same sign but larger magnitude.

7 Heterogeneous Effects & Robustness Checks

In this section, I estimate the benchmark model in two different subsamples. First, I remove "high-frequency" countries to estimate heterogeneous effects. Then, I remove China from the sample as a robustness check.

7.1 Heterogeneous Effects - "Low Frequency" Countries

It might be that a country that has a history of international disputes is impacted differently by having a new case than a country that is rarely involved in them.

I have data on cases since 1987 that provide information on the history of international disputes for the countries in the sample before the period covered by the FDI data. Using this, I classify countries as "high frequency" if they have more cases than the sample average (9 cases), and "low frequency" otherwise.

The "high frequency" developing economies are Argentina, Czech Republic, Ecuador, Egypt, Kazakhstan, Mexico, Ukraine, and Venezuela. I estimate the benchmark model from equation 2 without them in the sample, thus only for the "low frequency" countries. The estimation results are in Table A3. In Panel A of Table 6, I present the average treatment effects on the treated with the overall impact for this subsample.

In column 1, we see the coefficients for FDI flows one year after the country pair has a case. The sum of all the statistically significant coefficients is 1.724, representing a semi-elasticity of 461%. This implies an economic impact of 137 million dollars, significantly larger than the benchmark model effect of 9 million dollars.

In column 2, the effects for FDI flows two years after the country pair has a case are similar to the one-year lag. The sum of coefficients is 1.687, representing a semi-elasticity of 440%, and an economic effect of 131 million dollars. These results are also significantly larger than the benchmark model effect of 39 million dollars. Thus, when considering flows, countries in the "low-frequency" subsample receive larger FDI inflows than the overall sample one and two years after having a case.

In column 3, the impact on bilateral FDI positions one year after a case shows no statistically significant effects. In this specification, countries on the "low-frequency" subsample are not impacted either positively by the presence of a BIT or negatively due to a case.

Finally, column 4 presents the results on FDI positions two years after having a case. The total coefficient is 0.274, with a semi-elasticity of 32%, and an economic impact of

52 million dollars. These results are slightly larger than the 49 million dollars from the benchmark model. This suggests that for the more stable specification of FDI positions two years after a case, the effects for the "low-frequency" and "high-frequency" countries are similar.

7.2 Robustness Check - Sample without China

A relevant characteristic of the FDI data is its skewness. This reflects that a few countries receive most of the investment, in my sample, this is the case of China.

If I remove China from the sample, the mean of positive FDI bilateral positions drops from 166 to 108 million dollars. More impressively, the maximum value for FDI positions is 17% the maximum when including China. Thus, it might be the case that the extraordinary level of investment received by this country is driving the main results.

As a robustness check, I repeat the estimations from the benchmark model in equation 2 removing China from the sample. The results for all estimations are in Table A4. In Panel B of Table 6, I present the average treatment effects on the treated with the overall impacts.

Column 1 in Panel B displays the effects on FDI flows one year after a case for the benchmark model estimations in the sample without China. The total coefficient sum of 0.497 converts into a semi-elasticity of 64%, with an economic impact of 19 million dollars. These results are larger than the overall sample, 9 million dollars, though the magnitude is not as large as in Panel A for the "low-frequency" countries.

Meanwhile, I do not find statistically significant effects for this subsample in FDI flows two years after the case.

Moving to positions one year after a case, in column 2, I find negative effects. The sum of the statistically significant coefficients is -0.458. This represents a semi-elasticity of -37%, an economic impact of -11 million dollars. This is a change from the benchmark model where the result for this specification is 6 million dollars. It is of the same sign, though smaller in magnitude, than the extensions of the model presented in Table 5. Hence, it might be the case that a large positive China effect is preventing the results of the benchmark model from being negative. This suggests that overall, the effects on FDI positions one year after the dispute are negative, with varying magnitude.

Finally, the results for FDI positions two years after facing a dispute are in column 4.

The positive effects have an overall coefficient of 0.120, which implies a semi-elasticity of 13% and an economic valuation of 21 million dollars. These effects are smaller than the benchmark effects of 49 million dollars and are aligned with the overall stability of the results for this specification.

8 Summary and Concluding Remarks

Developing economies aim to attract foreign direct investment to overcome their lack of local domestic capital and promote growth. Institutional barriers that prevent an efficient international capital allocation are a vast and ever-growing interest in the literature.

In this paper, I approach this topic using Investor-State Dispute Settlement (ISDS), an institutional arrangement present in bilateral investment treaties (BITs), as a mechanism that reveals a developing economy reputation towards foreign direct investment. I focus on developing economies since they are expected to be more beneficiated by these international commitments devices that signal a stable institutional environment for foreign investors.

ISDS is expected to be an impartial international committee providing an objective decision about a dispute between a country and a foreign direct investor, thus operating as a mechanism where institutional quality is revealed. If this is true, having a case would imply a negative reputational shock that dampens future FDI in the developing economy of the destination. On the other hand, BITs work as a commitment device and should have a positive effect on attracting FDI. Since most of the time, these two institutional arrangements go hand in hand, their overall effect must be captured together. That is the main argument of this paper.

Findings in my benchmark model suggest that the overall effects of these international arrangements are positive and increase FDI flows and positions in developing economies. There is variation in magnitudes, which are larger two years after a case, particularly for positions.

When considering the composition of these overall positive effects, I find a negative reputational shock of 21 and 27 million dollars in FDI inflows one and two years after the case respectively. This is more than compensated by the positive effect of the interaction between a BIT and a case of 95 and 192 million dollars for each specification. Similarly, the negative effect of a case in FDI positions is about 65 million dollars while the interaction effect is 117 million dollars with a year lag and 185 with a two-year lag.

The first extension to the model includes the effects of losing the case. These results show a puzzling positive effect the first year after a case with an increase in FDI flow of 203 million dollars, the largest of all estimations. Nevertheless, in line with previous findings in the literature, I find a negative economic impact when I consider FDI flows two years after the case, and FDI positions one year later. With a two-year lag, FDI bilateral positions are again positive and of similar magnitude to those in the benchmark model.

The interaction of international and domestic institutions shows an effect similar to losing a case, but different in magnitude. The main difference is the estimation of FDI inflows one year after a case, which in this case is similar to the benchmark model. This extension yields the largest effect for FDI positions two years after a case. The complementarity effect is thus more relevant in this specification, which is positive in all models.

The analysis of heterogeneous effects shows positive economic impacts in FDI flows and positions larger for the subsample of countries that do not have a history of international disputes than for the whole sample. Furthermore, I do not find negative effects in any of the specifications, though my estimations for positions with a one-year lag are not statistically significant. This suggests there is no reputational shock for countries that do not have a large history of international arbitration with firms. Hence, dispute history matters for investors, as captured in the investor survey on the perception of ISDS mentioned before.

As a robustness check, I estimate the benchmark model removing China as an economy of destination, since it is the largest FDI recipient in the sample. Similar to the benchmark model, the effects on flow one year after a case is positive, though larger in magnitude. I do not find statistically significant effects for inflows in the two-year lag specification. The negative effects in positions one year after the case for this subsample, as well as for the model extensions, suggest the reason estimations for the benchmark model are positive is a large positive China effect. Finally, the results for positions two years after the case are positive, though smaller than in all previous estimations.

When considering the composition of these general effects, the individual coefficient for BIT only yields statistically significant results in some of the specifications. Thus, I do not find strong evidence that a bilateral investment treaty by itself attracts FDI in developing economies. The case coefficient is negative and statistically significant in all specifications except for the "low-frequency" subsample. This reflects the reputational cost of facing an international dispute, mostly driven by countries with a large history of cases. Meanwhile, the joint effect of having a case and a BIT is positive, suggesting that both mechanisms grant investors a reliable institutional environment in developing economies.

In sum, the negative reputational effect associated with an ISDS case is larger than the commitment device effect of a BIT for FDI inflows two years after the developing economy of the destination loses the case and when it interacts with the quality of the domestic institutions. For FDI positions, this negative effect occurs one year after the case in all but the benchmark model. These results are mostly driven by countries with a large history of international disputes showing that the mere presence of a case does not provide a bad reputation but rather a large track of them. In all models, the negative results disappear two years after the dispute when considering the more stable measure of FDI bilateral positions.

This topic is part of a wide area of research that interacts with academic fields outside of economics, making it very rich. Further research should try to incorporate another set of costs that ISDS has for countries. In particular, I do not consider the cost of awards, which have been increasing later making a case by legal authors against "crippling compensations". Similarly, I do not include the "regulatory chill" effects by which some governments stop implementing domestic regulations to prevent disputes.

The results from this paper could inform the current efforts to reform ISDS that are taking place at UNCITRAL and help develop a mechanism that is useful for firms and states.

References

Aisbett, Emma, Matthias Busse, and Peter Nunnenkamp, "Bilateral investment treaties as deterrents of host-country discretion: the impact of investor-state disputes on foreign direct investment in developing countries," *Review of World Economics*, February 2018, *154* (1), 119–155.

Alfaro, Laura, Sebnem Kalemli-Ozcan, and Vadym Volosovych, "Why doesn't capital flow from rich to poor countries? An empirical investigation," *The Review of Economics and Statistics*, 2008, p. 22.

Allee, Todd and Clint Peinhardt, "Contingent Credibility: The Impact of Investment Treaty Violations on Foreign Direct Investment," *International Organization*, July 2011, *65* (3), 401–432.

Anderson, James E and Eric Van Wincoop, "Gravity with Gravitas: A Solution to the Border Puzzle," *American Economic Review*, 2003, *93*, 24.

Baier, Scott L. and Jeffrey H. Bergstrand, "Do free trade agreements actually increase members' international trade?," *Journal of International Economics*, March 2007, *71* (1), 72–95.

_ , _ , and Matthew W. Clance, "Heterogeneous effects of economic integration agreements," *Journal of Development Economics*, November 2018, *135*, 587–608.

Berger, Axel, Matthias Busse, Peter Nunnenkamp, and Martin Roy, "Do trade and investment agreements lead to more FDI? Accounting for key provisions inside the black box," *International Economics and Economic Policy*, June 2013, *10* (2), 247–275.

Correia, Sergio, Paulo Guimarães, and Thomas Zylkin, "ppmlhdfe: Fast Poisson Estimation with High-Dimensional Fixed Effects," *arXiv:1903.01690 [econ]*, August 2019. arXiv: 1903.01690.

Fally, Thibault, "Structural Gravity and Fixed Effects," *Journal of International Economics*, 2015, p. 27.

Falvey, Rod and Neil Foster-McGregor, "Heterogeneous effects of bilateral investment treaties," *Review of World Economics*, November 2017, *153* (4), 631–656.

Green, Andrew, "Tabaré Vázquez," *The Lancet*, February 2021, *397* (10273), 468.

Head, Keith and John Ries, "FDI as an outcome of the market for corporate control: Theory and evidence," *Journal of International Economics*, January 2008, *74* (1), 2–20.

— **and Thierry Mayer**, "The United States of Europe: A Gravity Model Evaluation of the Four Freedoms," *Journal of Economic Perspectives*, May 2021, *35* (2), 23–48.

IMF, "Coordinated Direct Investment Survey Guide," Technical Report, Internatioinal Monetary Fund, Washington, D.C. 2015.
International Investment Agreements Navigator | UNCTAD Investment Policy Hub

***International Investment Agreements Navigator | UNCTAD Investment Policy Hub**, 2021.*

Lucas, Robert E., *"Why doesn't capital flow from rich to poor countries?," The American Economic Review, May 1990, 80 (2), 92–96.*

Ossa, Ralph, Robert W Staiger, and Alan O Sykes, *"Disputes in International Investment and Trade," 2020, p. 53.*

Queen Mary University of London, School of International Arbitration and Counsel International Arbitration Group, *"QMUL-CCIAG Survey: Investors' Perceptions of ISDS," 2020.*

Silva, J.M.C. Santos and Silvana Tanreyro, *"The Log of Gravity," The Review of Economics and Statistics, November 2006, 88 (4), 641–658.*

Stahler, Frank, *"An optimal investor-state dispute settlement mechanism," Review of International Economics, January 2021.*

Sykes, Alan O., *"Public vs. Private Enforcement of International Economic Law: Of Standing and Remedy," SSRN Electronic Journal, 2005.*

Tobin, Jennifer L. and Susan Rose-Ackerman, *"When BITs have some bite: The political-economic environment for bilateral investment treaties," The Review of International Organizations, March 2011, 6 (1), 1–32.*

Wellhausen, Rachel L., *"Investor–state disputes: when can governments break contracts?," Journal of Conflict Resolution, 2015, 59 (2), 239–261. Publisher: Sage Publications Sage CA: Los Angeles, CA.*

WorldBank, *"Doing Business," 2020.*

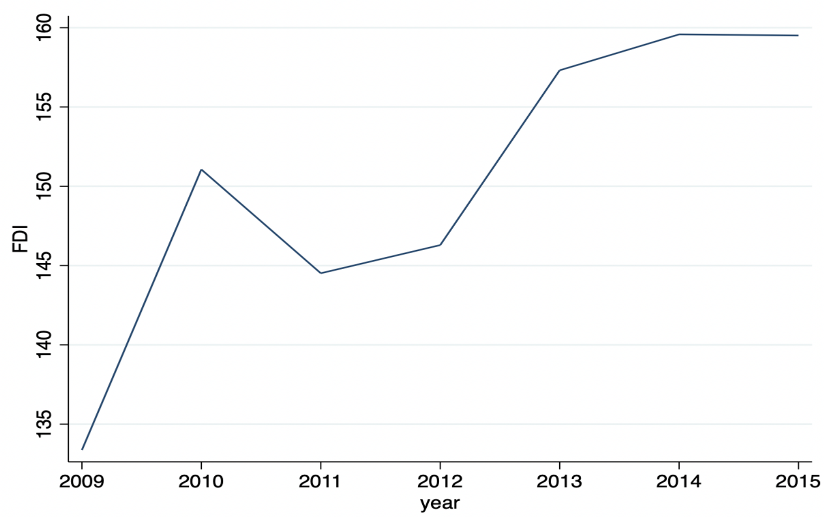
— , *"World Development Indicators | DataBank," 2021.*

Table 1. Distribution of Cases Registered in 2019 by Region

Region	Percentage
East Europe & Central Asia	25
South America	21
Middle East & North Africa	17
Sub-Saharan Africa	11
Western Europe	10
Central America & the Caribbean	8
South & East Asia & the Pacific	6
North America	2

Source: 2019 ICSID Annual Report

Figure 1. FDI inward positions, average by year
(USD millions)



NOTE: This figure plots the average by year of FDI inward positions in the sample for 2009-2015.

Table 2. Top 10 developing economies with more BITs in force (2015)

Country	Number of BITs
China	140
Egypt	114
Romania	88
Turkey	86
India	78
Ukraine	74
Bulgaria	72
Malaysia	71
Indonesia	66
Belarus	66

NOTE: This table shows the number of BITs in force for the 10 developing economies in the sample with more BITs in 2015.

Table 3. Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
FDI bilateral inward position*	60,339	165.65	4,154	0	390,522
FDI inflow*	40,330	29.77	469.67	0	55,774
BIT in force	139,216	0.15	0.36	0	1
Case	139,216	0.00	0.03	0	1
Investor	139,216	0.00	0.02	0	1
Investor-winning sector	139,216	0.00	0.02	0	1
Domestic	139,216	0.34	0.48	0	1

NOTE: *Current million dollars This table shows the number of observations, mean, standard deviation, maximum, and minimum values for all the variables used in the estimation. The period covered is from 2009 to 2015. Countries included are those not classified by the World Bank as "high income" in 2009.

Table 4. Reputational Shocks and Commitment Devices

Panel A: Estimation of coefficients (PPML)				
	(1) FDI flow	(2) FDI flow	(3) FDI position	(4) FDI position
BIT	1.108	1.141*	-0.383	-0.382
	(0.689)	(0.673)	(0.261)	(0.256)
Country pair has a case (t-1)	-1.180*		-0.498*	
	(0.645)		(0.266)	
BIT× Country pair has a case (t-1)	1.435**		0.536*	
	(0.724)		(0.315)	
Country pair has a case (t-2)		-2.320**		-0.490*
		(0.990)		(0.191)
BIT × Country pair has a case (t-2)		2.010*		0.749*
		(1.026)		(0.306)
Observations	10786	10786	20704	20704
Pseudo R-squared	0.965	0.965	0.991	0.991
All models include: Constant, country of origin-year FE, country of destination-year FE, and country pair FE.				
Panel B: Average Treatment Effects on the Treated				
Sum of significant coefficients	0.255	0.831	0.038	0.259
Semi-elasticities (%)	29	130	4	30
Economic impact (million USD)	9	39	6	49

NOTE: Panel A of this table shows the results of the PPML estimations based on equation 2. The dependent variable in the first two columns is the FDI flow in the developing economy of the destination from the economy of origin. In columns 3 and 4 the dependent variable is FDI positions. In columns 1 and 3, *Country pair has a case (t-1)* equals to one all the years after the economy of destination faces an ISDS case from a firm from the country of origin, lagged one year. In columns 2 and 4, *Country pair has a case (t-2)* considers two lagged years. *BIT* equals one if a BIT is in force between the country pair and zero otherwise. Standard errors (clustered by country pair) are in parentheses. *, **, *** represent statistical significance at the 10%, 5% and 1% level, respectively. Panel B shows the Average Treatment Effects on the Treated (ATT). It includes the sum of the estimated PPML coefficients statistically different from zero in Panel A and its conversion into semi-elasticities computed as $(e^{\hat{\beta}_i} - 1)$*100. The economic impact is evaluated at the sample mean FDI flows (USD 29.77 million) or positions (USD 165.65 million).

Table 5. Extensions: Losing a Case & Domestic Institutions

Average Treatment Effect on the Treated	(1) FDI flow (case t-1)	(2) FDI flow (case t-2)	(3) FDI position (case t-1)	(4) FDI position (case t-2)
Panel A: **Effects of Losing a Case**				
Sum of Coefficients	2.055	-0.423	-0.503	0.250
Semi Elasticities (%)	681	-34	-40	28
Economic Impact (million USD)	*203*	*-10*	*-65*	*47*
Panel B: **Domestic Institutions Complementarity**				
Sum of Coefficients	0.234	-2.338	-0.729	0.453
Semi Elasticities (%)	26	-90	-52	57
Economic Impact (million USD)	*8*	*-27*	*-86*	*95*
Benchmark Economic Impact	*9*	*39*	*6*	*49*

NOTE: This table shows the Average Treatment Effects on the Treated (ATT) for the estimations of equation 3 in Panel A, and equation 4 in Panel B. Estimation results are displayed in Table A1 and Table A2 respectively. It includes the sum of the estimated PPML coefficients statistically different from zero and its conversion into semi-elasticities computed as $(e^{\hat{\beta}_i}-1)*100$. The economic impact is evaluated at the sample mean of FDI flows (USD 29.77 million) or positions (USD 165.65 million).

Table 6. "Low Frequency" Countries & Sample without China

Average Treatment Effect on the Treated	(1) FDI flow (case t-1)	(2) FDI flow (case t-2)	(3) FDI position (case t-1)	(4) FDI position (case t-2)
Panel A: **"Low Frequency" Countries**				
Sum of Coefficients	1.724	1.687		0.274
Semi Elasticities (%)	461	440		32
Economic Impact (million USD)	*137*	*131*	*0*	*52*
Panel B: **Sample without China**				
Sum of Coefficients	0.497		-0.458	0.120
Semi Elasticities (%)	64		-37	13
Economic Impact (million USD)	*19*	*0*	*-11*	*21*
Benchmark Economic Impact	*9*	*39*	*6*	*49*

NOTE: This table shows the Average Treatment Effects on the Treated (ATT) for the estimations of equation 2. In Panel A the estimation sample only includes countries with less than nine cases ("Low Frequency"), and in Panel B the estimations are for the sample without China. Estimation results are displayed in Table A3 and Table A4 respectively. It includes the sum of the estimated PPML coefficients statistically different from zero and its conversion into semi-elasticities computed as $(e^{\hat{\beta}_i} - 1)$*100. The economic impact is evaluated at the sample mean of FDI flows (USD 29.77 million) or positions (USD 165.65 million).

A Appendix

A.1 Full Estimations for Losing a Case

In Table A1 I present the full set of coefficients for the effects of losing a case estimated from equation 3.

The effect of having a bilateral investment treaty between the country pair is only statistically significant when considering FDI flows two years after a case. Meanwhile, having a case is negative and statistically significant in all specifications, with columns 1 and 2 referring to a one-year lag, and columns 3 and 4 for two years. The interaction of BIT and case is positive in all cases, but only statistically significant in columns 1 and 4.

The variable indicating if the investor won the case is not statistically significant in any specification. It has a statistically significant and negative effect on flows two years after the case when interacted with the existence of a BIT. On the other hand, the interaction of having a case and an investor winning it is positive for flows one and two years after the case, and not statistically significant for positions.

Thus, the sole fact of an investor winning a case does not provide enough information for a reputational effect. It is the interaction of the whole set of variables and the institutional devices they represent that provide a clearer picture of the developing economy of a destination's reputation.

A.2 Full Estimations for Domestic Institutions Complementarity

In Table A2 I present the full set of coefficients for the effects of losing a case estimated from equation 4.

The effect of having a bilateral investment treaty between the country pair is only statistically significant when considering FDI positions one year after a case. Meanwhile, having a case is negative and statistically significant in all specifications, with columns 1 and 2 referring to a one-year lag, and columns 3 and 4 for two years. The interaction of BIT and case is positive in all cases, but not statistically significant in column 3.

The variable *Domestic*, which indicates if the quality of the domestic institutions in enforcing contracts is above the sample mean plus one standard deviation, is not statistically significant in its interaction with the BIT indicator. When interacting with a case, it is only statistically significant, and negative, on positions one year after the case. Meanwhile, the interaction of having a case and an investor winning it is positive for flows one and two years after the case, and not statistically significant for positions. The interaction of *Domestic* and having a case is only statistically significant for FDI positions one year after the case.

Thus, the sole fact of an investor winning a case does not provide enough information for a reputational effect. It is the interaction of the whole set of variables and the institutional devices they represent that provide a clearer picture of the developing economy of a destination's reputation.

Table A1. Effects of Losing a Case on FDI Bilateral Flows and Positions

	(1) FDI flow	(2) FDI flow	(3) FDI position	(41) FDI position
BIT	1.126	1.208*	-0.382	-0.380
	(0.701)	(0.670)	(0.262)	(0.256)
Country pair has a case (t-1)	-1.157*		-0.503*	
	(0.660)		(0.265)	
BIT × Country pair has a case (t-1)	1.378*		0.475	
	(0.736)		(0.315)	
Investor	1.581	2.089	-0.586	-0.092
	(1.581)	(1.384)	(0.479)	(0.541)
BIT × Investor	-1.972	-2.708*	0.111	-0.146
	(1.296)	(1.522)	(0.273)	(0.304)
Country pair has a case (t-1) × Investor	1.834*		0.453	
	(1.067)		(0.430)	
Country pair has a case (t-2)		-1.418*		-0.498***
		(0.729)		(0.193)
BIT × Country pair has a case (t-2)		0.689		0.748**
		(0.791)		(0.322)
Country pair has a case (t-2) × Investor		2.495***		0.162
		(0.926)		(0.493)
Observations	10786	10786	20704	20704
Pseudo R-squared	0.965	0.966	0.991	0.991
All models include: Constant, country of origin-year FE, country of destination-year FE, and country pair FE.				

NOTE: This table shows the results of the PPML estimations based on equation 3. The dependent variable in the first two columns is the FDI flow in the developing economy of the destination from the economy of origin. In columns 3 and 4 the dependent variable is FDI positions. In columns 1 and 3, *Country pair has a case (t-1)* equals to one all the years after the economy of destination faces an ISDS case from a firm from the country of origin, lagged one year. In columns 2 and 4, *Country pair has a case (t-2)* considers two lagged years. *BIT* equals one if a BIT is in force between the country pair and zero otherwise. The variable *Investor* equals 1 if the result of the dispute is favorable to the investor and zero otherwise. Standard errors (clustered by country pair) are in parentheses. *, **, *** represent statistical significance at the 10%, 5% and 1% level, respectively.

Table A2. Effects of Domestic Institutions Complementarity on FDI bilateral Flows and Positions

	(1) FDI flow	(2) FDI flow	(3) FDI position	(4) FDI position
BIT	1.027	1.099	-0.522*	-0.489
	(0.718)	(0.696)	(0.307)	(0.305)
Country pair has a case (t-1)	-1.196*		-0.487*	
	(0.657)		(0.264)	
BIT × Country pair has a case (t-1)	1.430*		0.780**	
	(0.735)		(0.303)	
BIT × Domestic	0.128	0.062	0.187	0.146
	(0.355)	(0.354)	(0.131)	(0.130)
Country pair has a case (t-1) × Domestic	0.119		-0.500**	
	(0.369)		(0.217)	
Country pair has a case (t-2)		-2.338**		-0.481**
		(0.984)		(0.194)
BIT × Country pair has a case (t-2)		1.461		0.934***
		(1.044)		(0.336)
Country pair has a case (t-2) × Domestic		0.707		-0.307
		(0.493)		(0.187)
Observations	10786	10786	20704	20704
Pseudo R-squared	0.965	0.965	0.991	0.991

All models include: Constant, country of origin-year FE, country of destination-year FE, and country pair FE.

NOTE: This table shows the results of the PPML estimations based on equation 4. The dependent variable in the first two columns is the FDI flow in the developing economy of the destination from the economy of origin. In columns 3 and 4 the dependent variable is FDI positions. In columns 1 and 3, *Country pair has a case (t-1)* equals to one all the years after the economy of destination faces an ISDS case from a firm from the country of origin, lagged one year. In columns 2 and 4, *Country pair has a case (t-2)* considers two lagged years. *BIT* equals one if a BIT is in force between the country pair and zero otherwise. The variable *Domestic* equals 1 if the destination country has a score equal to or larger than the sample mean plus one standard deviation of the *Score-enforcing contracts* variable from the WorldBank (2020), and zero otherwise. Standard errors (clustered by country pair) are in parentheses. *, **, *** represent statistical significance at the 10%, 5% and 1% level, respectively.

A.3 Full Estimations for "Low Frequency Countries

In Table A3 I present the full set of coefficients for the benchmark model in equarion 2 after removing from the sample countries with more than nine cases.

The effect of having a bilateral investment treaty between the country pair is highly statistically significant for FDI flows one and two years after a case (columns 1 and 2). Meanwhile, having a case is not statistically significant with a one-year lag, but is negative and statistically significant two years after (columns 2 and 4). The interaction of BIT and case yields positive and statistically significant results at the 10% level two years after a case both for flows (column 2) and positions (column 4).

When considering positions, I do not find statistically significant results with a one-year lag after a case. I only find negative statistically significant results for the effects of having a case two years after the case.

These results show that countries in the "Low Frequency" subsample experience the negative reputational effect of a case with a two-year lag.

A.4 Full Estimations for Sample without China

In Table A4 I present the full set of coefficients for the benchmark model in equation 2 after removing from China from the sample.

The BIT coefficient is not statistically significant in any of the specifications. The case coefficient is negative and statistically significant for a one-year lag both for flows, shown in column 1, and positions, in column 3. When interacting with the BIT variable, the coefficient is only significant for flows (column 1).

Two years after the dispute, I do not find effects for flows. For positions, I find a negative and highly statistically significant coefficient for the case variable. These effects get reverted in the interaction with a BIT and the statistically significant coefficient becomes positive.

This reflects that even after removing China from the sample, I still find the same overall results of a negative reputational effect for a case that is compensated in the interaction with a BIT.

Table A3. Heterogeneous Effects - "Low Frequency" Countries

	(1) FDI flow	(2) FDI flow	(3) FDI position	(4) FDI position
BIT	1.724***	1.723***	-0.373	-0.375
	(0.646)	(0.645)	(0.275)	(0.268)
Country pair has a case (t-1)	-1.065		-0.422	
	(0.715)		(0.319)	
BIT × Country pair has a case (t-1)	1.293		0.302	
	(0.796)		(0.372)	
Country pair has a case (t-2)		-2.301**		-0.481**
		(1.171)		(0.214)
BIT × Country pair has a case (t-2)		2.265*		0.755*
		(1.218)		(0.412)
Observations	9888	9888	19076	19076
Pseudo R-squared	0.968	0.968	0.991	0.991
All models include: Constant, country of origin-year FE, country of destination-year FE, and country pair FE.				

NOTE: This table shows the results of the PPML estimations based on equation 2 after removing "High Frequency" countries of destination from the sample. These countries have more cases than the average from 1987 to 2015 (more than nine cases). The dependent variable in the first two columns is the FDI flow in the developing economy of the destination from the economy of origin. For columns 3 and 4 it is FDI positions. In columns 1 and 3, *Country pair has a case (t-1)* equals to one all the years after the developing economy of destination faces an ISDS case from a firm from the country of origin lagged one period. In columns 2 and 4, *Country pair has a case (t-2)* considers two lagged years. *BIT* equals one if a BIT is in force between the country pair. Standard errors (clustered by country pair) are in parentheses. *, **, *** represent statistical significance at the 10%, 5% and 1% level, respectively.

Table A4. Robustness Check - Sample without China

	(1) FDI flow	(2) FDI flow	(3) FDI position	(4) FDI position
BIT	0.023	0.033	0.201	0.200
	(1.201)	(1.183)	(0.287)	(0.289)
Country pair has a case (t-1)	-1.342**		-0.458*	
	(0.617)		(0.273)	
BIT × Country pair has a case (t-1)	1.839***		0.525	
	(0.704)		(0.331)	
Country pair has a case (t-2)		-1.114		-0.485***
		(0.689)		(0.164)
BIT × Country pair has a case (t-2)		1.008		0.605***
		(0.786)		(0.234)
Observations	10466	10466	20170	20170
Pseudo R-squared	0.965	0.965	0.986	0.986
All models include: Constant, country of origin-year FE, country of destination-year FE, and country pair FE.				

NOTE: This table shows the results of the PPML estimations based on equation 2 after removing from the sample China as a country of destination. The dependent variable in the first two columns is the FDI flow in the developing economy of the destination from the economy of origin. For columns 3 and 4 it is FDI positions. In columns 1 and 3, *Country pair has a case (t-1)* equals to one all the years after the developing economy of destination faces an ISDS case from a firm from the country of origin lagged one period. In columns 2 and 4, *Country pair has a case (t-2)* considers two lagged years. *BIT* equals one if a BIT is in force between the country pair. Standard errors (clustered by country pair) are in parentheses. *, **, *** represent statistical significance at the 10%, 5% and 1% level, respectively.

Table A5. Cases by year, country pair, and income level

Year	Country of destination	Country of origin	BIT	Destination Income	Origin Income
2009	Argentina	Spain	Yes	Upper Medium	High
2009	Argentina	United Kingdom	Yes	Upper Medium	High
2009	Belize	United Kingdom	Yes	Lower Medium	High
2009	Costa Rica	Germany	Yes	Upper Medium	High
2009	Ecuador	United States of America	Yes	Lower Medium	High
2009	Egypt	United States of America	Yes	Lower Medium	High
2009	Guatemala	Spain	Yes	Lower Medium	High
2009	Kazakhstan	Netherlands	Yes	Upper Medium	High
2009	Cambodia	United States of America	No	Low	High
2009	Sri Lanka	Germany	Yes	Lower Medium	High
2009	Moldova	Russia	Yes	Lower Medium	Upper Medium
2009	Mexico	Spain	Yes	Upper Medium	High
2009	North Macedonia	Switzerland	Yes	Upper Medium	High
2009	El Salvador	United States of America	Yes	Lower Medium	High
2009	Ukraine	United States of America	Yes	Lower Medium	High
2009	Venezuela	Canada	Yes	Upper Medium	High
2010	Bulgaria	Germany	Yes	Upper Medium	High
2010	Belize	United Kingdom	Yes	Lower Medium	High
2010	Bolivia	United Kingdom	Yes	Lower Medium	High
2010	Guatemala	United States of America	No	Lower Medium	High
2010	India	Australia	Yes	Lower Medium	High
2010	Kazakhstan	Moldova	No	Upper Medium	Lower Medium
2010	Kazakhstan	Netherlands	Yes	Upper Medium	High
2010	Kazakhstan	United States of America	Yes	Upper Medium	High
2010	Lithuania	Italy	Yes	Upper Medium	High
2010	Peru	Argentina	Yes	Upper Medium	Upper Medium
2010	Peru	France	Yes	Upper Medium	High
2010	Romania	Turkey	Yes	Upper Medium	Upper Medium
2010	Romania	United States of America	Yes	Upper Medium	High

Table A5 – Cases by year, country pair, and income level

Year	Country of destination	Country of origin	BIT	Destination Income	Origin Income
2010	Turkmenistan	Turkey	Yes	Lower Medium	Upper Medium
2010	Tanzania	United Kingdom	Yes	Low	High
2010	Tanzania	Hong Kong	No	Low	High
2010	Uruguay	Switzerland	Yes	Upper Medium	High
2010	Uzbekistan	Israel	Yes	Lower Medium	High
2010	Venezuela	Barbados	Yes	Upper Medium	High
2010	Venezuela	Switzerland	Yes	Upper Medium	High
2010	Venezuela	Spain	Yes	Upper Medium	High
2010	Zimbabwe	Germany	Yes	Low	High
2011	Albania	Greece	Yes	Lower Medium	High
2011	Albania	Italy	Yes	Lower Medium	High
2011	Ecuador	Canada	Yes	Upper Medium	High
2011	Ecuador	United States of America	Yes	Upper Medium	High
2011	Indonesia	United Kingdom	Yes	Lower Medium	High
2011	Indonesia	Saudi Arabia	Yes	Lower Medium	High
2011	Kyrgyzstan	Latvia	Yes	Low	Upper Medium
2011	Libya	Kuwait	No	Upper Medium	High
2011	Moldova	France	Yes	Lower Medium	High
2011	Mongolia	Canada	Yes	Lower Medium	High
2011	Peru	France	Yes	Upper Medium	High
2011	Peru	United States of America	No	Upper Medium	High
2011	Philippines	Germany	Yes	Lower Medium	High
2011	Turkmenistan	United Kingdom	Yes	Upper Medium	High
2011	Turkey	Netherlands	Yes	Upper Medium	High
2011	Uzbekistan	United Kingdom	Yes	Lower Medium	High
2011	Venezuela	Barbados	Yes	Upper Medium	High
2011	Venezuela	Canada	Yes	Upper Medium	High
2011	Venezuela	Switzerland	Yes	Upper Medium	High
2011	Venezuela	Netherlands	Yes	Upper Medium	High

Table A5 – Cases by year, country pair, and income level

Year	Country of destination	Country of origin	BIT	Destination Income	Origin Income
2011	Venezuela	Portugal	Yes	Upper Medium	High
2012	Costa Rica	Spain	Yes	Upper Medium	High
2012	Algeria	Luxembourg	Yes	Upper Medium	High
2012	Egypt	United States of America	Yes	Lower Medium	High
2012	Guinea	France	Yes	Low	High
2012	Indonesia	United Kingdom	Yes	Lower Medium	High
2012	India	Mauritius	Yes	Lower Medium	Upper Medium
2012	Moldova	Russia	Yes	Lower Medium	High
2012	Moldova	Ukraine	Yes	Lower Medium	Lower Medium
2012	North Macedonia	Netherlands	Yes	Upper Medium	High
2012	Montenegro	Netherlands	Yes	Upper Medium	High
2012	Romania	Italy	Yes	Upper Medium	High
2012	Venezuela	Barbados	Yes	Upper Medium	High
2012	Venezuela	Canada	Yes	Upper Medium	High
2012	Venezuela	France	Yes	Upper Medium	High
2012	Venezuela	Netherlands	Yes	Upper Medium	High
2012	Venezuela	Portugal	Yes	Upper Medium	High
2013	Costa Rica	Switzerland	Yes	Upper Medium	High
2013	Jordan	Kuwait	Yes	Upper Medium	High
2013	Kazakhstan	United States of America	Yes	Upper Medium	High
2013	Kyrgyzstan	Canada	No	Lower Medium	High
2013	Pakistan	Turkey	Yes	Lower Medium	Upper Medium
2013	Panama	United States of America	Yes	Upper Medium	High
2013	Papua New Guinea	Singapore	No	Lower Medium	High
2013	Venezuela	Spain	Yes	Upper Medium	High
2014	Dominican Republic	United States of America	No	Upper Medium	High
2014	Montenegro	Cyprus	Yes	Upper Medium	High
2014	Peru	Canada	Yes	Upper Medium	High
2014	Turkey	France	Yes	Upper Medium	High

Table A5 – Cases by year, country pair, and income level

Year	Country of destination	Country of origin	BIT	Destination Income	Origin Income
2015	Cameroon	Luxembourg	Yes	Lower Medium	High
2015	Senegal	United Kingdom	Yes	Low	High
2015	Ukraine	United Kingdom	Yes	Lower Medium	High

NOTE: This table includes all cases by year, country-pair, and income level. Data source is Wellhausen (2015)

Table A6. Destination Economies, inward direct investment positions

Afghanistan	Egypt	Mayotte	Syria
Albania	El Salvador	Mexico	Tajikistan
Algeria	Eritrea	Micronesia	Tanzania
American Samoa	Eswatini	Moldova	Thailand
Angola	Ethiopia	Mongolia	Timor-Leste
Anguilla	Falkland Islands	Montenegro	Togo
Antigua & Barbuda	Fiji	Montserrat	Tokelau
Argentina	French South Terr.	Morocco	Tonga
Armenia	Gabon	Mozambique	Tunisia
Azerbaijan	Gambia	Myanmar	Turkey
Bangladesh	Georgia	Namibia	Turkmenistan
Belarus	Ghana	Nepal	Tuvalu
Belize	Grenada	Nicaragua	Uganda
Benin	Guadeloupe	Niger	Ukraine
Bhutan	Guatemala	Nigeria	Uruguay
Bolivia	Guinea	Niue	Uzbekistan
Bonaire, St. Eust. & Saba	Guinea-Bissau	Norfolk Island	Vanuatu
Bosnia & Herzegovina	Guyana	North Macedonia	Venezuela
Botswana	Haiti	Pakistan	Vietnam
Bouvet Island	Heard Is. & McDonald Is.	Palau	Wallis & Futuna
Brazil	Holy See	Panama	West Bank & Gaza
British Indian Oc. Terr.	Honduras	Papua New Guinea	Western Sahara
Bulgaria	India	Paraguay	Yemen
Burkina Faso	Indonesia	Peru	Zambia
Burundi	Iran	Philippines	Zimbabwe
Cabo Verde	Iraq	Pitcairn	Rwanda
Cambodia	Jamaica	Romania	South Africa
Cameroon	Jordan	Russia	Saudi Arabia
Central African Rep.	Kazakhstan	Senegal	Serbia
Chad	Kenya	Réunion	Seychelles

Table A6 – Destination Economies

Chile	Kiribati	St. Helena, Asc. & Tristan	Singapore
China	Korea (People's Rep.)	St. Kitts & Nevis	Slovakia
Christmas Islands	Kyrgyzstan	Saint Lucia	Slovenia
Cocos (Keeling) Is.	Lao	St. Vincent & Grenadines	Solomon Islands
Colombia	Lebanon	Samoa	South Africa
Comoros	Lesotho	Sao Tome & Principe	South Korea
Congo	Liberia	Senegal	Spain
Congo (Democratic Rep.)	Libya	Serbia	Sri Lanka
Cook Islands	Lithuania	Seychelles	Sweden
Costa Rica	Madagascar	Solomon Is.	Switzerland
Cuba	Malawi	Somalia	Tajikistan
Côte d'Ivoire	Malaysia	Tanzania	Thailand
Djibouti	Mali	S. Georgia & S.Sandwich	Togo
Dominica	Marshall Islands	South Sudan	Uganda
Dominica	Martinique	Sri Lanka	United Kingdom
Dominican Republic	Mauritania	Sudan	United States of Ame
Ecuador	Mauritius	Suriname	Uruguay
Albania	Côte d'Ivoire	Macao	Venezuela
Algeria	Denmark	Malaysia	West Bank & Gaza
Argentina	El Salvador	Mali	Zambia
Armenia	Estonia	Malta	Niger
Aruba	Eswatini	Mauritius	Turkey
Australia	Eswatini	Mexico	Ukraine
Austria	Finland	Moldova	
Azerbaijan	France	Moldova	
Azerbaijan	Georgia	Mongolia	
Bahrain	Germany	Montenegro	
Bangladesh	Ghana	Morocco	
Barbados	Greece	Mozambique	
Belarus	Guatemala	Myanmar	
Belgium	Guinea-Bissau	Namibia	
Benin	Honduras	Nepal	

Table A6 – Destination Economies

Bhutan	Hong Kong	Netherlands
Bolivia	Hungary	New Zealand
Bosnia and Herzegovina	Iceland	
Botswana	India	Nigeria
Brazil	Indonesia	North Macedonia
Bulgaria	Ireland	Norway
Burkina Faso	Israel	Pakistan
Cabo Verde	Italy	Palau
Cambodia	Japan	Panama
Canada	Jordan	Paraguay
Chile	Kazakhstan	Peru
China	Kazakhstan	Philippines
Costa Rica	Kuwait	Poland
Croatia	Kyrgyzstan	Portugal
Croatia	Latvia	Romania
Curaçao	Lebanon	Russia
Cyprus	Lithuania	Rwanda
Czech Republic	Luxembourg	Samoa

The Data source is the CDIS survey, IMF, for the period 2009-2015.